DELHI PUBLIC SCHOOL NEWTOWN
SESSION: 2022-23
HALF YEARLY EXAMINATION

CLASS:X | **FULL MARKS: 80**
SUBJECT: CHEMISTRY [SET A] | **TIME: 2 HOURS**

General Instructions:

- **The paper consists of six printed pages.**
- **Section A is compulsory. Attempt *any four* questions from Section B.**
- **Answers should be to the point.**
- **Question numbers should be copied carefully while answering the questions.**

SECTION A
(Attempt all questions from this section)

Question 1

Choose one correct answer to the questions from the given options: **[15]**

(i) The metal which reacts with concentrated sodium hydroxide to produce hydrogen gas is:
- A. Iron
- B. Aluminium
- C. Copper
- D. Gold

(ii) Electron affinity is highest for the element is:
- A. Group 1
- B. Group 2
- C. Group 17
- D. Group 18

(iii) In Hall Heroult's process,the component of the electrolytic mixture which is taken in higher ratio is:
- A. Cryolite
- B. Flurospar
- C. Alumina
- D. Powdered coke

(iv) The main component of brass is:
- A. Zinc
- B. Copper
- C. Tin
- D. Iron

(v) A polar covalent compound having two lone pairs:
- A. Methane
- B. Ammonia
- C. Water
- D. Carbon tetrachloride

(vi) An acid which has two replaceable hydrogen ions is:

- **A. Acetic acid**
- **B. Phosphorous acid**
- **C. Nitric acid**
- **D. Formic acid**

(vii) The hydroxide which is soluble in excess of ammonium hydroxide is:

- **A. Ferric hydroxide**
- **B. Lead hydroxide**
- **C. Copper hydroxide**
- **D. Calcium hydroxide**

(viii) The RMM of sulphur dioxide is 64,its vapour density is:

- **A. 16**
- **B. 32**
- **C. 48**
- **D. 64**

(ix) The gas evolved when concentrated hydrochloric acid is heated with manganese dioxide is:

- **A. Chlorine**
- **B. Hydrogen**
- **C. Hydrogen chloride**
- **D. Oxygen**

(x) The promoter used in Haber's process is:

- **A. Finely divided iron**
- **B. Nickel**
- **C. Platinum**
- **D. Molybdenum**

(xi) The gas released when copper turnings reacts with concentrated sulphuric acid is:

- **A. Hydrogen**
- **B. Hydrogen sulphide**
- **C. Sulphur dioxide**
- **D. Sulphur trioxide**

(xii) The akaline earth metal found in period 3 and group 2 is:

- **A. Boron**
- **B. Magnesium**
- **C. Calcium**
- **D. Barium**

(xiii) An aqueous compound which turns methyl orange yellow is:

- **A. Ammonium hydroxide**
- **B. Nitric acid**
- **C. Anhydrous calcium chloride**
- **D. Sulphuric acid**

(xiv) The acid with basicity two:

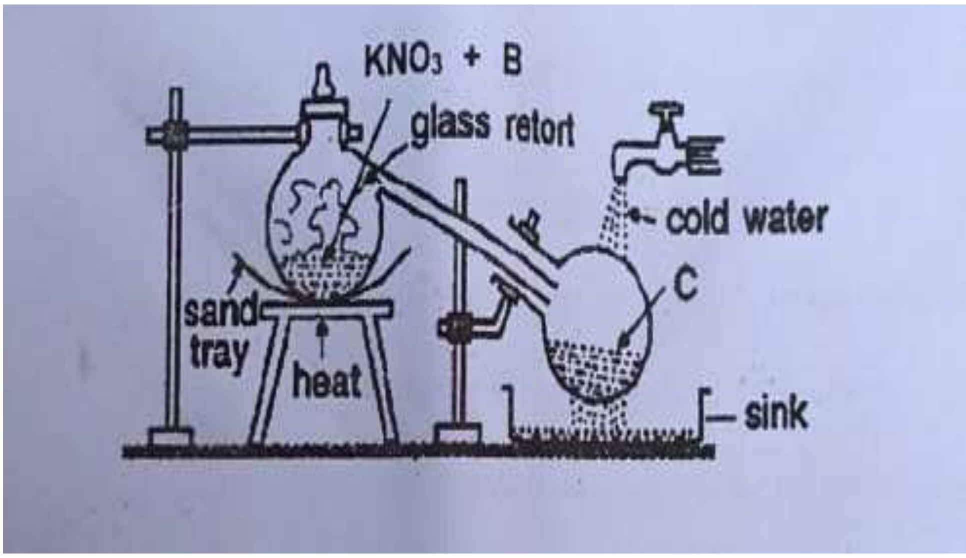
- **A. Phosphoric acid**
- **B. Sulphuric acid**
- **C. Nitric acid**
- **D. Acetic acid**

(xv) The nitrate which on heating leaves no residue is:

A. Ammonium nitrate
B. Sodium nitrate
C. Silver nitrate
D. Calcium nitrate

Question 2

(i) Observe the diagram given below which illustrates the laboratory set up for the preparation of an acid. Answer the question that follows: [5]



(a) Name the reactant B.
(b) Name the product C.
(c) What is the speciality of the apparatus? Give reason for the same.
(d) Why is cold water poured over the collection tank?

(ii) Match the following Column A with Column B: [5]

Column A	Column B
(a) Acid salt	1. Dirty green in colour
(b) Ferrous hydroxide	2. Chalky white
(c) Lead hydroxide	3. Carbon tetrachloride
(d) Copper metal	4. Potassium bicarbonate
(e) Non polar covalent compound	5. Reddish brown in colour

(iii) Complete the following by choosing the correct answer from the brackets: [5]

(a) Liquor ammonia is ____________ [neutral/ basic]
(b) Covalent compounds are insoluble in _________ [water/ organic solvents]
(c) The carbonate ore of zinc is _________ [Zincite/ calamine]
(d) The metal whose nitrate on heating produces oxygen as the only gaseous product ________ [sodium/ silver]
(e) Acidic gas gives dense white fumes of _________ [ammonium chloride/ammonium hydroxide] with ammonia.

(iv) Identify the following: **[5]**

(a) The gas which burns with a greenish yellow flame.

(b) A metal which produces a green solution on reacting with dilute sulphuric acid.

(c) The metal which imparts strength to duralumin.

(d) The element used as electrodes in electrolytic reduction of alumina.

(e) The gas which fumes in moist air.

(v) Correct the following sentences by adding suitable word/s: **[5]**

(a) Ammonia gas turns red litmus paper blue.

(b) Magnesium reacts with very dilute nitric acid to produce hydrogen.

(c) Sodium chloride is a conductor of electricity.

(d) The aim of the fountain experiment is to demonstrate solubility of hydrogen chloride in water.

(e) Carbon reacts with nitric acid to produce nitrogen dioxide.

SECTION B

(Attempt any four questions from this section)

Question 3

(i) Identify the Anions present in each of the following compounds: **[2]**

(a) To a solution of compound B equal volume of freshly prepared ferrous sulphate solution is added followed by addition of concentrated sulphuric acid dropwise along the wall of the test tube, a brown ring is formed at the junction of the two solutions.

(b) On addition of dilute sulphuric acid to compound D a colourless gas with rotten egg odour is produced which turns moist lead acetate paper silvery black.

(ii) Write the products and balance the following equations: **[2]**

(a) $Cu + \text{conc } HNO_3$

(b) $FeS + HCl \rightarrow$

(iii) Arrange the following as per the instruction given in the brackets: **[3]**

(a) Na, Cl, Si, S (decreasing order of electronegativity)

(b) C, B, N, O (increasing order of metallic character)

(c) Br, F, I, Cl (decreasing order of atomic radius)

(iv) Fill in the blanks selecting the appropriate word from the given choice: **[3]**

(a) In an ionic compound the bond is formed due to __________ of electrons. (sharing/ transfer)

(b) A molecule which has three covalent bonds ____________ (ammonia/ water)

(c) Electrovalent compounds do not conduct electricity in __________ state (molten/ solid)

Question 4

(i) For each of the substances given below, what is the role played in the extraction of Aluminium. **[2]**

(a) Fluorspar

(b) Powdered coke

(ii) Calculate: [3]

(a) Give the empirical formula of $C_6H_{12}O_6$

(b) A gas cylinder holds 100 g of hydrogen gas under certain conditions of temperature and pressure. If an identical cylinder with the same capacity can hold 400 g of gas 'P' under the same conditions of temperature and pressure, calculate the vapour density of gas 'P'.

(iii) The following questions are pertaining to the laboratory preparation of hydrogen chloride gas: [3]

(a) Why is the temperature kept below 200°C ?

(b) Why is concentrated sulphuric acid used for this process?

(c) How is the gas collected?

(iv) Explain the following: [2]

(a) Nitric acid cannot be concentrated beyond 68%.

(b) Excess of air is taken for the industrial preparation of nitric acid.

Question 5

(i) (a) Write an equation for the dissociation of ammonium chloride in aqueous medium.

(b) What colour fountain is seen for the fountain experiment for ammonia? [2]

(ii) Identify the cation present in the following compounds based on the observations:

(a) A pungent smelling gas is released when compound 'X' is heated with an alkali. The gas released turns alkaline potassium mercuric iodide solution brown.

(b) A reddish brown precipitate is obtained when sodium hydroxide is added to an aqueous solution of 'Y'. [2]

(iii) Give balanced chemical equation for the following: [3]

(a) Bauxite is reacted with hot concentrated sodium hydroxide.

(b) Copper turnings are reacted with dilute nitric acid.

(c) Dilute hydrochloric acid is added sodium sulphite.

(iv) State one relevant observation for each of the following reactions: [3]

(a) Excess of ammonia is passed through an aqueous solution of zinc nitrate.

(b) Concentrated nitric acid is added to carbon.

(c) Copper oxide is reacted with dilute sulphuric acid.

Question 6

(i) Define: [2]

(a) pH

(b) Gangue

(ii) Solve: [2]

Acetylene burns in air forming carbon dioxide and water vapour. Calculate the volume of oxygen needed and carbon dioxide formed on complete combustion of 50 cm^3 of acetylene.

$$2C_2H_2\,(g) + 5O_2\,(g) \rightarrow 4CO_2\,(g) + 2H_2O\,(l)$$

(iii) State the conditions required for the following reactions: [3]

(a) Conversion of ammonia to nitric oxide.

(b) Conversion of sulphur dioxide to sulphur trioxide.

(c) Conversion of nitrogen to ammonia.

(iv) Identify the chemical properties of the compounds underlined in the following reactions: [3]

(a) $MgSO_3 + 2\underline{HCl} \rightarrow MgCl_2 + H_2O + SO_2$

(b) $Zn + 2\underline{H_2SO_4}$ (conc) $\rightarrow ZnSO_4 + 2H_2O + SO_2$

(c) $2\underline{NH_3} + 3PbO \rightarrow 3Pb + 3H_2O + N_2$

Question 7

(i) Empirical formula of a compound is XY_2. If its empirical formula weight is equal to its vapour density, calculate the molecular formula of the compound. [2]

(ii) Identify the gas released in each of the following cases: [2]

(a) Iron pyrites is roasted.

(b) Excess of ammonia is reacted with chlorine.

(iii) A white crystalline solid M on heating decomposes with a crackling sound producing a reddish brown gas 'N', a colourless, odourless gas 'O' and leaves behind a residue 'P', which is yellow in both hot and cold condition. [3]

(a) Identify M.

(b) Give one chemical test for gas N.

(c) Write balanced chemical equation for action of heat on M.

(iv) Choose the answer from the list which fits the description: [3]

[CaO, ZnO, CO_2, NO_2, SiO_2, SO_2]

(a) A mixed acid anhydride

(b) An amphoteric oxide

(c) An oxide of a metalloid

Question 8

(i) Draw the electron dot and cross structure of the following: [2]

(a) Ammonium ion

(b) Water

(ii) Distinguish between the following pair of compounds: [2]

(a) Hydrochloric acid and sulphuric acid using lead nitrate solution.

(b) Sodium sulphite and sodium sulphate using dilute hydrochloric acid.

(iii) Differentiate between the following: [3]

(a) Sodium chloride and carbon tetrachloride [constituent particles]

(b) Alkali metals and Alkaline earth metals number of valence electrons]

(c) Hydracids and oxyacids [composition]

(iv) An element Z has atomic number 16. Answer the following questions: [3]

(a) State the period and the group to which it belongs.

(b) What is the valency of Z?

(c) Write the formula of the compound formed between hydrogen and Z.